

WEEK 3

SELLER AND INVENTORY MANAGEMENT SYSTEM

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Week 3 Task Report

Scenario

The e-commerce company sells products supplied by multiple sellers. The system should maintain seller details, track the products supplied by each seller, and monitor inventory levels to ensure product availability.

1. Create Seller and Inventory Tables

```
SQL> CREATE TABLE Seller(Seller_ID VARCHAR(7), Seller_Name VARCHAR(25), Email VARCHAR(30), Phone NUMBER(10), Address VARCHAR(45));
Table created.

SQL> CREATE TABLE Inventory(Inventory_ID VARCHAR(7), Product_ID VARCHAR(7), Seller_ID VARCHAR(7), Stock_Quantity NUMBER, Stock_status VARCHAR(11), Last_Updated DATE);
Table created.
```

2. Establish Relationships

Seller and Product

One seller can supply multiple products. One product can be supplied by one or multiple sellers.

Relationship: Seller -> Product is one to many.

Product and Inventory

Each product has inventory information. Inventory is used to track the available stock.

Relationship: Product -> Inventory is one to one.

Primary Key added for Both Tables

```
SQL> ALTER TABLE Seller ADD CONSTRAINT PK_Seller PRIMARY KEY(Seller_ID);
Table altered.

SQL> ALTER TABLE Inventory ADD CONSTRAINT PK_Inv PRIMARY KEY(Inventory_ID);
Table altered.
```

Add an appropriate primary key to both tables so that each record is uniquely identified.

Foreign Key for Inventory Table

The inventory table should contain a foreign-key reference to the related product, linking product information with its stock details.

```
SQL> ALTER TABLE Inventory ADD CONSTRAINT FK_Prod FOREIGN KEY(Product_ID) REFERENCES Product(Product_ID);
Table altered.

SQL> ALTER TABLE Inventory ADD CONSTRAINT FK_Seller FOREIGN KEY(Seller_ID) REFERENCES Seller(Seller_ID);
Table altered.
```

Constraints to make Unique Email ID

```
SQL> ALTER TABLE Seller ADD CONSTRAINT Un_mail UNIQUE(Email);
Table altered.
```

Apply a UNIQUE constraint to the seller email column so duplicate email IDs are not accepted.

To make Email NOT NULL

```
SQL> ALTER TABLE Seller MODIFY Email NOT NULL;
Table altered.
```

Add Constraint to Check the Stock Quantity is Positive

Use a CHECK constraint to ensure that stock quantity is not negative and only valid inventory values are entered.

```
SQL> ALTER TABLE Inventory ADD CONSTRAINT CK_Quan CHECK(Stock_Quantity >=0);
Table altered.
```

Insert Values to Seller Table

Insert sample seller records into the Seller table for testing and report generation.

```
SQL> INSERT INTO Seller VALUES('&SellID','&SellerName','&email','&Phone','&Address');
Enter value for sellid: S101
Enter value for sellername: Anuj Enterprises
Enter value for email: Anuj@gmail.com
Enter value for phone: 9840430082
Enter value for address: Madambakkam,Vengaivasal
old 1: INSERT INTO Seller VALUES('&SellID','&SellerName','&email','&Phone','&Address')
new 1: INSERT INTO Seller VALUES('S101','Anuj Enterprises','Anuj@gmail.com','9840430082','Madambakkam,Vengaivasal')

1 row created.
```

```
SQL> SELECT * FROM Seller;
```

SELLER_	SELLER_NAME	EMAIL	PHONE
ADDRESS			
S101	Anuj Enterprises Madambakkam,Vengaivasal	Anuj@gmail.com	9840430082
S102	Sanjay & CO Irumbuliyur,Tambaram	sanjay6385@gmail.com	6385102628
S103	Hema Enterprises Santhoshpuram,Tamilnadu	hema@gmail.com	9003197590
SELLER_	SELLER_NAME	EMAIL	PHONE
ADDRESS			
S104	Vanaja & CO pallikaranai,Vellacherry	vanaj@gmail.com	9710021922
S105	Deenu Enterprises Pareri,SP Kovil	shinchandeeps30@gmail.com	9043707830

Insert Values to Inventory Table

Insert sample inventory values associated with the products.

```
SQL> INSERT INTO Inventory VALUES('Inv109','P109','S101',40,'Available',TO_DATE('2026-09-17','YYYY-MM-DD'));
1 row created.

SQL> INSERT INTO Inventory VALUES('Inv101','P101','S101',100,'Available',TO_DATE('2026-09-13','YYYY-MM-DD'));
1 row created.

SQL> INSERT INTO Inventory VALUES('Inv102','P102','S101',50,'Available',TO_DATE('2026-09-13','YYYY-MM-DD'));
1 row created.

SQL> INSERT INTO Inventory VALUES('Inv103','P103','S101',75,'Available',TO_DATE('2026-09-14','YYYY-MM-DD'));
1 row created.

SQL> INSERT INTO Inventory VALUES('Inv104','P104','S101',80,'Available',TO_DATE('2026-09-14','YYYY-MM-DD'));
1 row created.
```

INVENTO	PRODUCT	SELLER_	STOCK_QUANTITY	STOCK_STATU	LAST_UPDA
-----	-----	-----	-----	-----	-----
Inv109	P109	S101	40	Available	17-SEP-26
Inv101	P101	S101	100	Available	13-SEP-26
Inv102	P102	S101	50	Available	13-SEP-26
Inv103	P103	S101	75	Available	14-SEP-26
Inv104	P104	S101	80	Available	14-SEP-26
Inv105	P105	S101	60	Available	15-SEP-26
Inv106	P106	S101	45	Available	15-SEP-26
Inv107	P107	S101	55	Available	16-SEP-26
Inv108	P108	S101	100	Available	16-SEP-26
Inv110	P110	S101	70	Available	17-SEP-26

Report 1: Seller-wise Product Report

Display:

- Seller name
- Number of products supplied
- Available stock

Use JOIN operations to get the required

```
SQL> SELECT s.Seller_Name, COUNT(i.Product_ID) AS Number_of_Products, SUM(i.Stock_Quantity) AS Available_Stock FROM Seller s JOIN Inventory i ON s.Seller_ID = i.Seller_ID WHERE i.Stock_status = 'Available' GROUP BY s.Seller_Name;
```

SELLER_NAME	NUMBER_OF_PRODUCTS	AVAILABLE_STOCK
-----	-----	-----
Hema Enterprises	1	3
Deenu Enterprises	1	3
Anuj Enterprises	10	675
Vanaja & CO	1	4
Sanjay & CO	1	3

Report 2: Stock Availability Report

Display:

- Product name
- Stock quantity
- Stock status

```
SQL> SELECT p.Product_Name, i.Stock_Quantity, i.Stock_Status
2 FROM Product p
3 JOIN Inventory i ON p.Product_ID = i.Product_ID;
```

```
PRODUCT_NAME
```

```
STOCK_QUANTITY STOCK_STATU
```

```
Metformin
40 Available
```

```
Paracetamol
100 Available
```

```
Amoxicillin
50 Available
```

```
PRODUCT_NAME
```

```
STOCK_QUANTITY STOCK_STATU
```

```
Cetirizine
75 Available
```

```
Omeprazole
80 Available
```

```
Vitamin C
60 Available
```

```
PRODUCT_NAME
```

```
STOCK_QUANTITY STOCK_STATU
```

```
Cough Syrup
45 Available
```

```
Antiseptic Cream
```

Report 3: Inventory Analysis

Find:

- Total products available
- Products out of stock
- Highest stocked products
- Average inventory quantity

```
SQL> SELECT COUNT(*) AS Total_Available_Products FROM Inventory WHERE Stock_Status = 'Available';
```

```
TOTAL_AVAILABLE_PRODUCTS
```

```
14
```

```
SQL> SELECT COUNT(*) AS OutOfStock_Products FROM Inventory WHERE Stock_Status = 'Out of Stock';
```

```
OUTOFSTOCK_PRODUCTS
```

```
0
```

```
SQL> SELECT p.Product_Name, i.Stock_Quantity
2  FROM Product p
3  JOIN Inventory i ON p.Product_ID = i.Product_ID
4  WHERE i.Stock_Quantity = (SELECT MAX(Stock_Quantity) FROM Inventory);
```

PRODUCT_NAME

STOCK_QUANTITY

Paracetamol
100

Bandage
100

```
SQL> SELECT AVG(Stock_Quantity) AS Average_Inventory
2  FROM Inventory;
```

AVERAGE_INVENTORY

49.1428571

END OF WEEK 3 REPORT